



A tool for designing highly efficient guide RNAs in CRISPR-Cas systems

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glossary

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) — a family of DNA sequences found in the genomes of prokaryotic organisms such as bacteria and archaea.

Guide RNA (gRNA) — the RNA component of the CRISPR-Cas9 genome engineering tool. Contains a ~20 base pair sequence that is complementary to the genomic target.

CRISPR-associated endonuclease protein (Cas) — a family of proteins able to cleave nucleic acids. Cas9 is the most commonly used Cas protein in CRISPR experiments.

Protospacer adjacent motif (PAM) — a short sequence of nucleotides that must be present downstream of the target site for the nuclease to make a double strand break.

GC-content (guanine-cytosine content) is the percentage of nitrogenous bases in a DNA or RNA molecule that are either guanine (G) or cytosine (C).

Indel — the insertion or deletion of nucleotides in the genome.

CRISPR-Cas system

A simple two-component system that can be used to target a specific genomic sequence and then make a cut in the DNA.

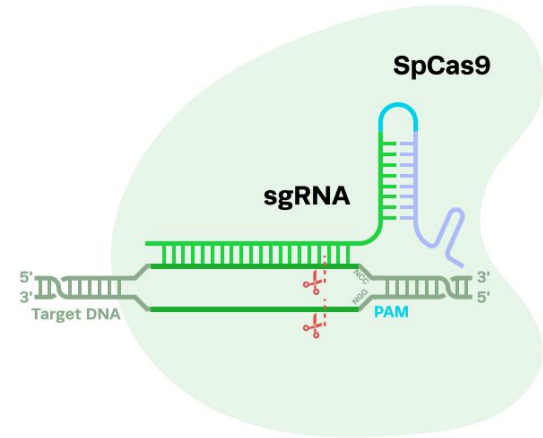
Components:

- a guide RNA (**gRNA**) that is specific to the target DNA sequence
- a CRISPR-associated endonuclease (**Cas**)

Protospacer adjacent motif (**PAM**): a short sequence of nucleotides that must be present downstream of the target site

Process:

- gRNA guides the system to a specific DNA site
- Cas protein cuts DNA in the right place



Minimum required components of a CRISPR/Cas system

uses of CRISPR-Cas9 and its modifications

Knock-Out

Making a gene permanently inoperative

Knock-in

Integrating a foreign genetic sequence into a cell's genome

Gene Visualization

Tagging desired parts of the genome



Gene expression regulation

Activating or silencing gene expression without changing the DNA sequence using inactive Cas9

Base Editing

precise, single nucleotide substitutions in a DNA or RNA strand without DSBs

application areas

Research

- Gene screening
- Creating cells with desired mutations

Biotechnologies

- Improving the properties of agricultural animals and plants
- Control of the spread of animal-borne infections
- Optimization of microorganisms for bio-production

Therapy

- Treatment of genetic diseases (editing cells outside the body)
- Potential for cancer therapy

gRNA efficiency

What can go wrong:

The weak connection between gRNA and target site – the protein does not have enough time to make the cleavage

Binding to an undesirable DNA region (off-target effect)

Weak stability of the system – early disintegration of the complex

Some properties affecting the gRNA efficiency:

sequence length, GC-content, specific nucleotides at specific positions, DNA-gRNA hybridization free energy, RNA free folding energy

existing tools for gRNA efficiency prediction

Property	Commercial products	Scientific research
Open source		✓
Availability of training datasets		✓
Convenient interface	✓	
Domestic product		

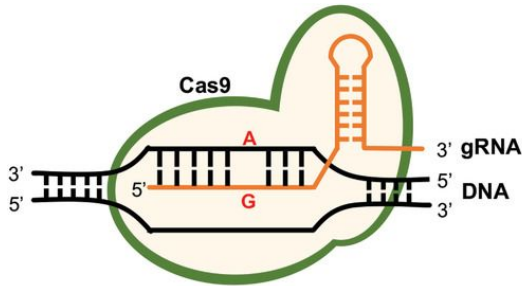
Multiple sgRNAs with three mismatches could achieve the 80% of the relative activity of the perfect matched sgRNA.

Meanwhile, it could reduce the number of off-target sites using sgRNAs with on-target mismatches

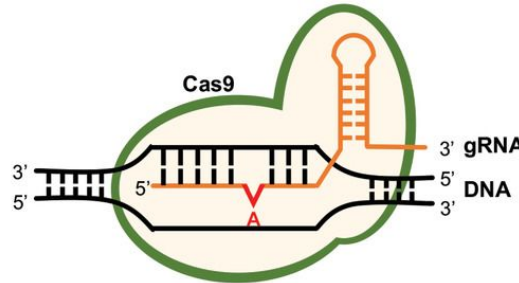
Product idea: Creating a domestic RNA design tool with a user-friendly web interface

Problem: Current popular sgRNA design tools only consider the sgRNA perfectly matched to the target site and provide the results without any on-target mismatch.

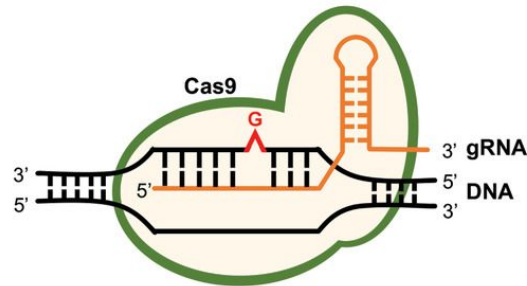
types of off-target effects



(a) Mismatch



(b) RNA bulge (insertion)



(c) DNA bulge (deletion)

a) off-target sites with base mismatch

b) off-target site with missing base (RNA bulge or insertion)

c) off-target sites with extra base (DNA bulge or deletion)

The cases (b) and (c) are considered as the indel (insertion or deletion) off-target events.

Y. Lin, T.J. Cradick et al. (2014) CRISPR/Cas9 systems have off-target activity with insertions or deletions between target DNA and guide RNA sequences

non-sequence information

- position on target in a gene
- thermodynamic properties
- local chromatin structure
- concentration of Cas9-gRNA complex
- different cell lines
- position of target sites with respect to transcription start site (TSS)
- target in exon

+ improved model accuracy

+ optimization for specific organisms

- limitation of the versatility of models

- complexity and heterogeneity of data

- lack of data for training

The use of non-sequence information is a compromise between accuracy and versatility of the model. **As for now, we will not use it.**

These data improve predictions in specific conditions, but limit the application of the model in a broader context.

X. Wu et al. (2014) Target specificity of the CRISPR-Cas9 system

V. Konstantakos et al. (2022) CRISPR-Cas9 gRNA efficiency prediction: an overview of predictive tools and the role of deep learning

mismatched data

GSE132080

4 898 sgRNAs targeting **2 449** genes, with growth phenotypes in K562 and Jurkat cells

~120 000 sgRNA variants, harboring 1-2 mismatch to original sgRNAs

Dataset¹ contains perfectly matched sgRNA (PMsgRNA), mismatched sgRNA (MMsgRNA), and the activity of MMsgRNA relative to PMsgRNA

Other datasets with mismatched gRNAs: K562², HEK293t², II4³, II5³, II6³

¹ M. Jost et al. (2020) *Titration of gene expression using libraries of systematically attenuated CRISPR guide RNAs*

² J. Lin et al. (2020) *CRISPR-net: a recurrent convolutional network quantifies CRISPR off-target activities with mismatches and indels*

³ G. Chuai et al. (2018) *DeepCRISPR: optimized CRISPR guide RNA design by deep learning*

data preprocessing

- Developed a database
- Added 5 new features, including encoding DNA-RNA pairs (see additional materials at the end of presentation)
- Conducted EDA

Code and description are available at [GitHub](#)



datasets for model validation

After training, we will predict values for benchmark datasets and compare the performance of our model with existing ones

Some of the datasets for validation: **CRISPOR**, **GUIDE-seq**

Dataset name	Number of instances	Species/Cell line
<i>Labuhn</i>	424	Human/HEL
<i>Shalem</i>	1278	Human/A375
<i>Koike-Yusa</i>	1064	Mouse/mESC
<i>Xi Xiang</i>	10 592	Human/HEK293T
<i>Shkumatava</i>	162	Zebrafish
<i>Gagnon</i>	111	Zebrafish

V. Konstantakos et al. (2022) CRISPR-Cas9 gRNA efficiency prediction: an overview of predictive tools and the role of deep learning

models

name	year	aim	architecture	additional features
CRISPRon	2021	efficiency prediction	CNN	
CRISPR-BERT	2024	off-target	Inception-based CNN, a BERT-base module and a BiGRU	adaptive batch-wise class balancing strategy for imbalanced off-target data
ExsgRNA	2022	generating a a gRNA with 1-3 mismatches	seq2seq-attention model. Two double layers, 256 units long short-term memory (LSTM) as encoder and decoder. Long attention mechanism with beam width=4	

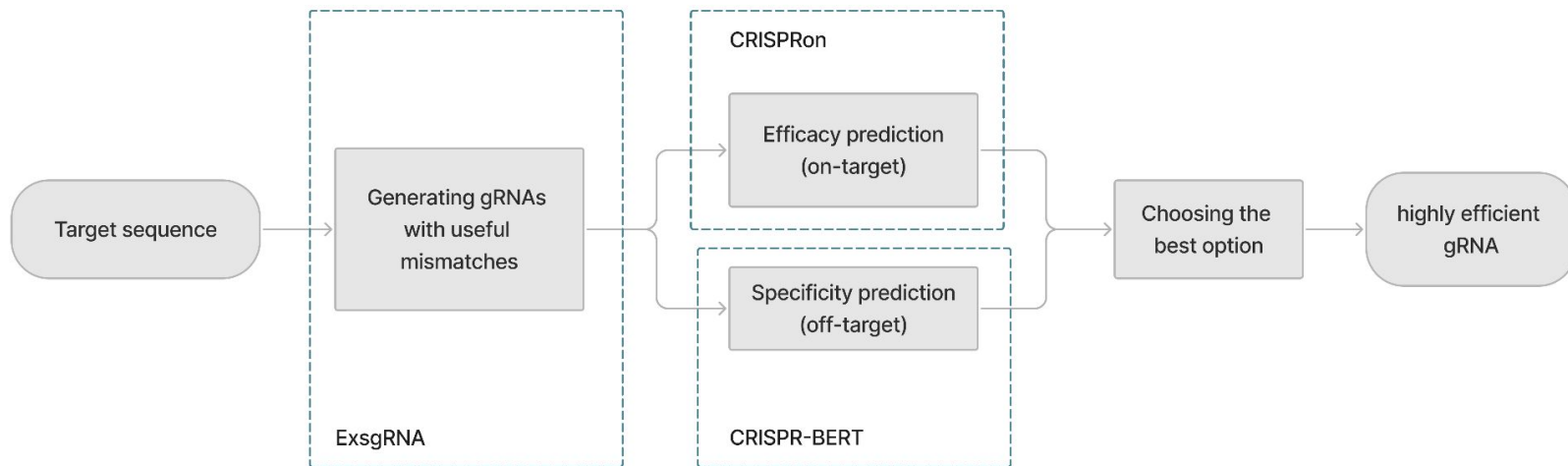
X. Xiang et al. (2021) Enhancing CRISPR-Cas9 gRNA efficiency prediction by data integration and deep learning

Y. Luo et al. (2024) Interpretable CRISPR/Cas9 off-target activities with mismatches and indels prediction using BERT

W.X. Hu et al. (2022) ExsgRNA: reduce off-target efficiency by on-target mismatched sgRNA

pipeline

We plan to dive into Specificity prediction stage to enhance existing solutions.
See the following slide for the ways of improvement



relevance

- **Limited existing models:**
Only two models (ExsgRNA and CRISPR-BERT) use datasets with mismatches/indels for training.
- **Use of advanced language models:**
Explore cutting-edge, open-source transformers (e.g., GPT-2, RoBERTa, ELECTRA)
- **Alternative encoding schemes:**
Experiment with different sequence representations (see additional slides).
- **Novel embedding techniques:**
Use innovative embedding tools for better guide RNA prediction.
- **Transfer learning strategies:**
Apply pre-trained models to adapt to specific guide RNA design challenges

results

- Studied the fundamental principles and mechanisms of CRISPR-Cas systems to understand the scientific and technical context of the project.
- Conducted literature review
- Identified gaps in existing tools and propose two product ideas: A pipeline integrating existing models. A novel approach using state-of-the-art architectures not previously applied to this task.
- Justified the proposed solutions by evaluating their scientific relevance
- Compared functionality and architecture of existing gRNA efficiency prediction models
- Found and preprocessed a dataset suitable for training predictive and/or generative models.
- Identified datasets suitable for model validation

roadmap

[illegible]

resources

Articles

- Motoche-Monar, C.; Ordoñez, J.E.; Chang, O.; Gonzales-Zubiate, F.A. gRNA Design: How Its Evolution Impacted on CRISPR/Cas9 systems Refinement. *Biomolecules* 2023, 13, 1698. [https:// doi.org/10.3390/biom13121698](https://doi.org/10.3390/biom13121698)
- Redman M, King A, Watson C, King D. What is CRISPR/Cas9? *Arch Dis Child Educ Pract Ed*. 2016 Aug;101(4):213-5. doi: 10.1136/archdischild-2016-310459. Epub 2016 Apr 8. PMID: 27059283; PMCID: PMC4975809
- Corsi, G.I., Qu, K., Alkan, F. et al. CRISPR/Cas9 gRNA activity depends on free energy changes and on the target PAM context. *Nat Commun* 13, 3006 (2022). <https://doi.org/10.1038/s41467-022-30515-0>
- Haeussler, M., Schönig, K., Eckert, H. et al. Evaluation of off-target and on-target scoring algorithms and integration into the guide RNA selection tool CRISPOR. *Genome Biol* 17, 148 (2016). <https://doi.org/10.1186/s13059-016-1012-2>
- Bhat, A.A., Nisar, S., Mukherjee, S. et al. Integration of CRISPR/Cas9 with artificial intelligence for improved cancer therapeutics. *J Transl Med* 20, 534 (2022). <https://doi.org/10.1186/s12967-022-03765-1>
- Vasileios Konstantakos, Anastasios Nentidis, Anastasia Krithara, Georgios Paliouras, CRISPR–Cas9 gRNA efficiency prediction: an overview of predictive tools and the role of deep learning, *Nucleic Acids Research*, Volume 50, Issue 7, 22 April 2022, Pages 3616–3637, <https://doi.org/10.1093/nar/gkac192>
- Zhang, G.; Xie, H.; Dai, X. DeepIndel: An Interpretable Deep Learning Approach for Predicting CRISPR/Cas9-Mediated Editing Outcomes. *Int. J. Mol. Sci.* 2024, 25, 10928. <https://doi.org/10.3390/ijms252010928>

Other

- Просто о сложном: CRISPR/Cas | [Биомолекула](#)
- CRISPR 101: You Guide To Understanding CRISPR | Synthego
- CRISPR: Guide to gRNA design | Snapgene Academy

THANKS!

I will be glad to answer your questions

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Кодирование пар РНК-ДНК

ИЛИ для несовпадений

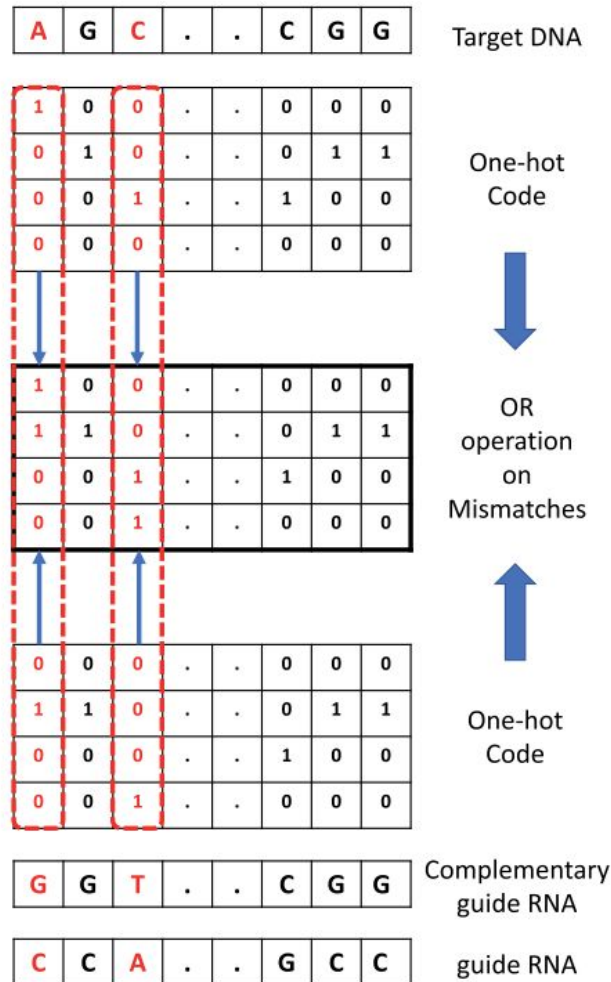
РНК и ДНК кодируются с помощью one-hot encoding как матрицы размера $4 \times N$, где N – длина последовательности. Затем матрицы объединяются с помощью операции ИЛИ

Представление: матрица $4 \times N$

Недостатки:

- пара АА будет закодирована также, как А
- потеря информации – нуклеотид из РНК или ДНК?
- только несовпадения

J. Lin, K.C. Wong (2018) Off-target predictions in CRISPR-Cas9 gene editing using deep learning



Кодирование пар РНК-ДНК

Конкатенация матриц

РНК и ДНК кодируются с помощью one-hot encoding как матрицы размера $4 \times N$, где N – длина последовательности. Затем матрицы конкатенируются

Представление: матрица $8 \times N$

Недостатки:

- только несовпадения

J. Charlier et al. (2021) Accurate deep learning off-target prediction with novel sgRNA-DNA sequence encoding in CRISPR-Cas9 gene editing

G	G	G	...	T	G	G
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Target DNA

0	0	0	...	0	0	0
1	1	1	...	0	1	1
0	0	0	...	0	0	0
0	0	0	...	1	0	0

One-Hot Encoding
of the Nucleotides



0	0	0	...	0	0	0
1	1	1	...	0	1	1
0	0	0	...	0	0	0
0	0	0	...	1	0	0
0	1	0	...	0	0	0
0	0	1	...	0	1	1
0	0	0	...	0	0	0
1	0	0	...	1	0	0

$8 \times L$ Encoded
Matrix
with sgRNA-DNA
Information



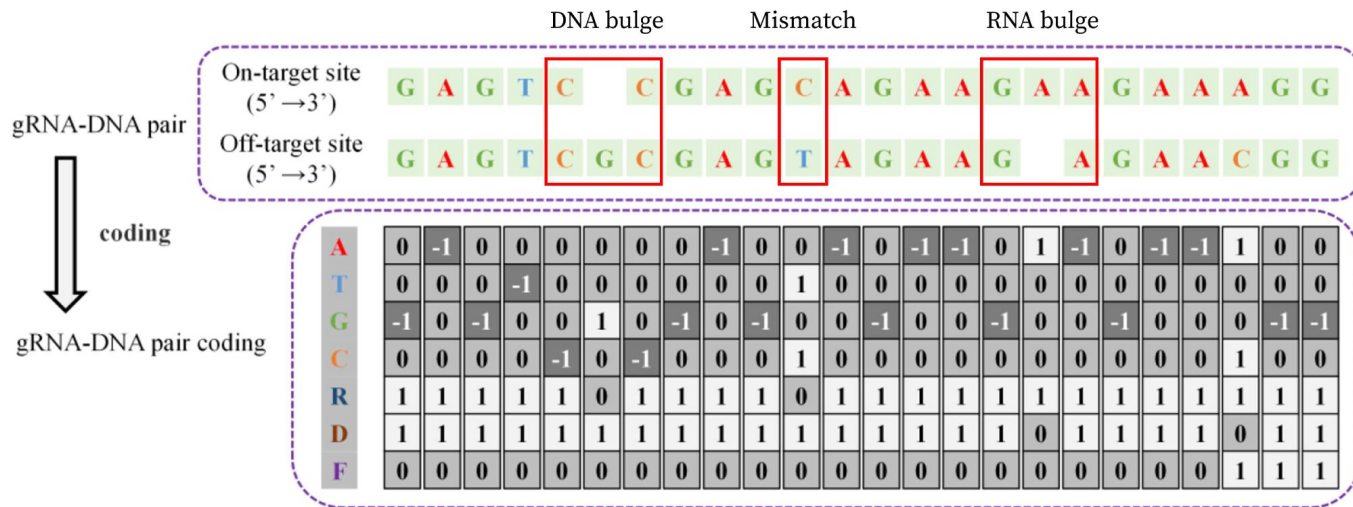
0	1	0	...	0	0	0
0	0	1	...	0	1	1
0	0	0	...	0	0	0
1	0	0	...	1	0	0

One-Hot Encoding
of the Nucleotides

T	A	G	...	T	G	G
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Guide RNA

Кодирование пар РНК-ДНК



Представление: матрица 7 x N

Благодаря трем дополнительным каналам сохраняется информация о:

- типе расхождения (Mismatch, RNA/DNA bulge)
- присутствии нуклеотида в ДНК или РНК
- функционале (gRNA или PAM)

Z.R. Zhang, Z.R. Jiang (2022) Effective use of sequence information to predict CRISPR-Cas9 off-target